

To package the **Ghost** program (Marvin) while keeping all dependencies for a backup, the best practice for Raspberry Pi and Linux environments is to create a **Requirements Manifest** combined with a **Tarball Archive**.

Because Python virtual environments (`venv`) use **hardcoded absolute paths**, you cannot simply copy the `marvin_env` folder to a new location or machine and expect it to work. Instead, you should follow this professional backup strategy:

1. Create a Dependency Manifest (Inside `ghost` directory)

First, "freeze" all the libraries Marvin uses into a text file. This ensures you have an exact record of the versions used (like `llama-cpp-python`, `pygame`, etc.).

```
bash

cd /home/samantha/ghost
source marvin_env/bin/activate
pip freeze > requirements.txt
deactivate
```

2. Archive the Core System (The "Tarball")

Now, create a compressed backup of your code, models, and configuration files. We will **exclude** the `marvin_env` folder because it's easier and safer to recreate it from the `requirements.txt` than to try and fix broken paths in a moved environment.

Run this from your home directory:

```
bash

cd /home/samantha
tar -czvf marvin_backup_$(date +%F).tar.gz --exclude='ghost/marvin_env' ghost/
```

- **-czvf**: Create, Gzip, Verbose, File.
- **--exclude='ghost/marvin_env'**: Keeps the backup small by leaving out the replaceable virtual environment.
- **\$(date +%F)**: Automatically adds today's date to the filename.

3. How to Restore from Backup

If you ever need to restore Marvin on a fresh system or different directory:

1. Extract the files:

```
bash

tar -xzf marvin_backup_YYYY-MM-DD.tar.gz
cd ghost
```

Rebuild the Brain's Environment:

```
bash
```

```
python3 -m venv marvin_env
source marvin_env/bin/activate
pip install -r requirements.txt
```

Summary Table for Backup Components

Component	Status	Backup Method
Logic (.py files)	Essential	Included in Tarball
Brain (brain.gguf)	Heavy	Included in Tarball (ensure it's in /models)
Protocol Registry	Dynamic	Included in Tarball (or auto-generated on boot)
Dependencies	Volatile	Saved via <code>pip freeze > requirements.txt</code>

By keeping the `requirements.txt` and the `models/` folder together in your tarball, you have a **100% portable copy** of Marvin that can be redeployed on any Raspberry Pi 5 in minutes.